NE 511Spring 2026

Homework 1 **Due 02/12/2026**

Problem 1 (35 points)

For a LWR at the beginning of cycle (BOC) assume that 90% of power comes from fission of U-235 and 10% comes from fission of Pu-239 built in the burnt fuel assemblies. At the end of cycle (EOC) this ratio changes to 70% for U-235 and 30% for Pu-239. Consider two additional states during the cycle: MOC1) 85% for U-235 and 15% for Pu-239; and MOC2) 80% for U-235 and 20% for Pu-239. Calculate the core average value for physical delayed neutron fraction at BOC, MOC1, MOC2, and EOC and determine % change.

For the same LWR, the prompt neutron lifetime is 10^{-4} s. We define the delayed neutron lifetime per delayed neutron group as the time interval between the fission and the time the delayed neutrons are absorbed or leaked. Calculate the corresponding core average neutron lifetime taking into account both prompt and delayed neutrons.

Delayed neutron group	*Decay constants (sec^{-1})*	*Delayed neutron fractions of U-235*	*Delayed neutron Fractions of Pu-239*
1	0.012375	0.00021	0.00007
2	0.030130	0.00142	0.00063
3	0.111774	0.00128	0.00044
4	0.301304	0.00257	0.00069
5	1.136065	0.00075	0.00018
6	3.013043	0.00027	0.00009
Total β		0.00650	0.00210
Total ν		2.43	2.90

Problem 2 (30 points)

Given the balance equation for the precursor density,

$$\frac{dc_k}{dt} = -\lambda_k c_k + S_{Pk} = -\lambda_k c_k + \sum_{i=1}^{N} \Sigma_{fi} \phi \nu_{dki} \quad \text{with } k = 1, \ldots, 6,$$

demonstrate the complications arising from the isotope dependence of the precursor decay constants.

Problem 3 (35 points)

Estimate the fractions of prompt and delayed neutrons that can cause fast fission in ^{238}U, assuming a sharp threshold at 1 MeV. Use emission spectra given in the table below.

Hint: Apply linear interpolation in the group around 1 MeV.

Group Number	u_g	E_g (eV)	χ_1	χ_2	χ_3	χ_4, χ_5, χ_6	χ
_	0	10 x 10^6	_	_	_	_	_
1	0.5	6.065 x 10^6	_	_	_	_	0.0325
2	1.0	3.679 x 10^6	_	_	_	_	0.1217
3	1.5	2.231 x 10^6	_	_	_	_	0.2109
4	2.0	1.353 x 10^6	_	_	_	0.009	0.2230
5	2.5	0.821 x 10^6	0.009	0.022	0.012	0.025	0.1728
6	3.0	0.498 x 10^6	0.021	0.066	0.070	0.062	0.1105
7	3.5	0.302 x 10^6	0.093	0.295	0.191	0.184	0.0628
8	4.0	0.183 x 10^6	0.088	0.110	0.094	0.128	0.0316
9	4.5	0.111 x 10^6	0.156	0.098	0.097	0.157	0.0168
10	5.0	0.674 x 10^5	0.174	0.108	0.156	0.109	0.0083
11	5.5	0.409 x 10^5	0.171	0.107	0.142	0.109	0.0040
12	6.0	0.248 x 10^5	0.131	0.088	0.118	0.099	0.0019
13	6.5	0.150 x 10^5	0.121	0.079	0.080	0.089	0.0009
14	7.0	0.912 x 10^4	0.020	0.013	0.015	0.015	0.0004
15	7.5	0.553 x 10^4	0.010	0.009	0.012	0.010	0.0002
16	8.0	0.335 x 10^4	0.005	0.004	0.005	0.004	0.0001
17	8.5	0.203 x 10^4	0.001	0.001	_	_	_
18	9.0	0.123 x 10^4	_	_	_	_	_
19	9.5	749	_	_	_	_	_
20	10.0	454	_	_	_	_	_
21	10.5	275	_	_	_	_	_
22	11.0	167	_	_	_	_	_
23	11.5	101	_	_	_	_	_
24	12 0	61.4	_	_	_	_	_
25	12.5	37.3	_	_	_	_	_
26	13.0	22.6	_	_	_	_	_